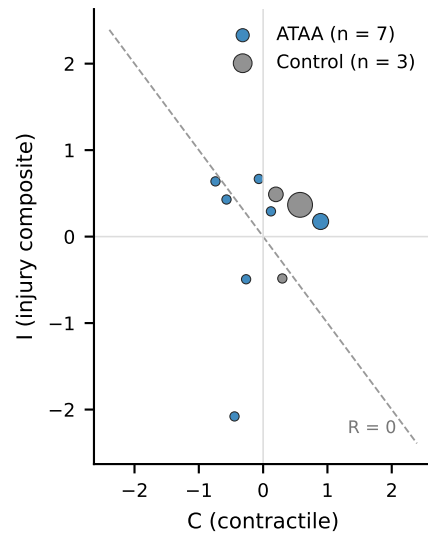
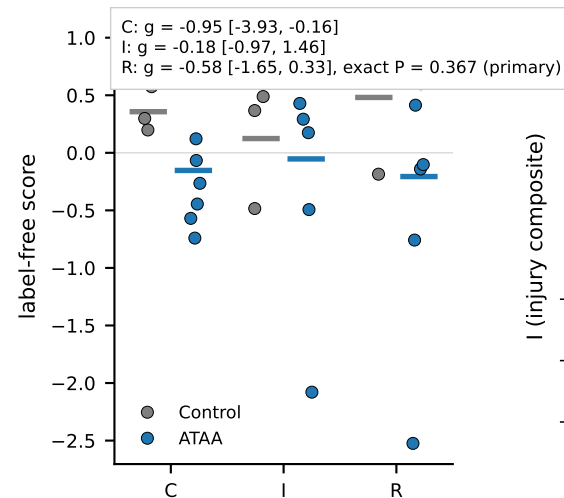


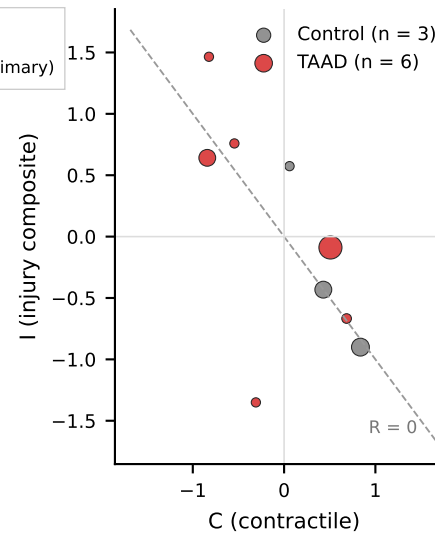
a GSE155468 (ATAA): per-patient C vs I, contractile SMC
(point size = number of cells)



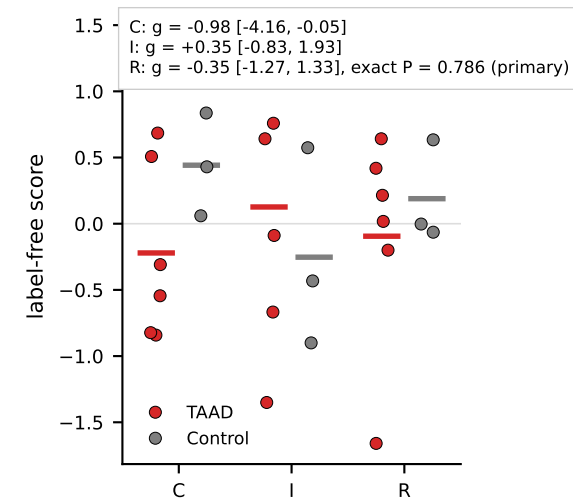
b GSE155468: ATAA vs control, contractile SMC



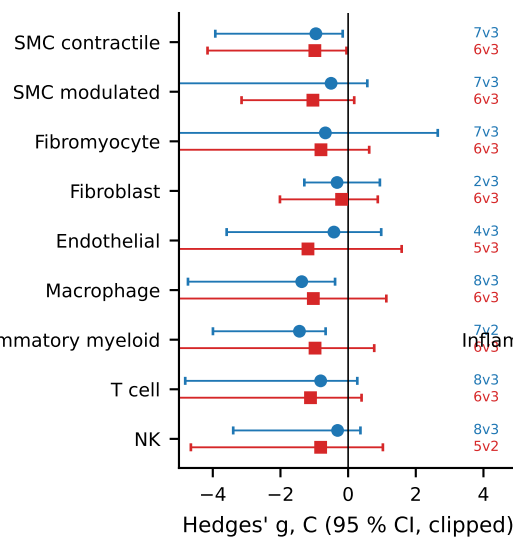
c GSE213740 (TAAD): per-patient C vs I, contractile SMC
(point size = number of cells)



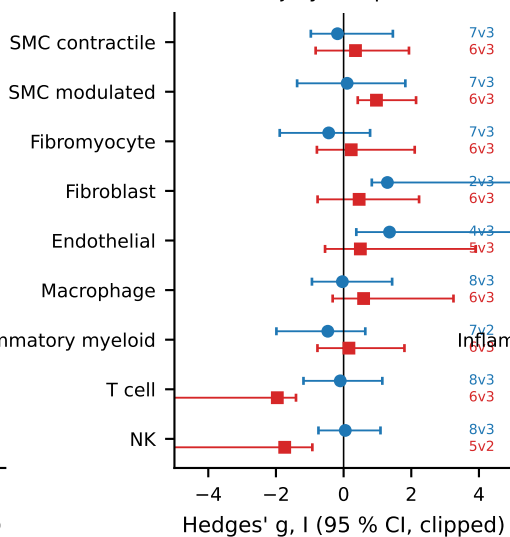
d GSE213740: TAAD vs control, contractile SMC



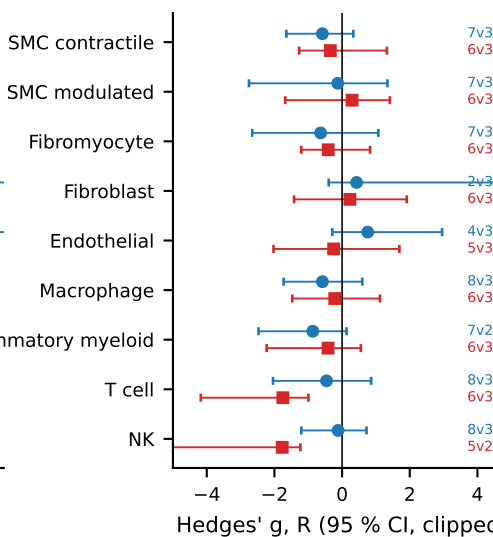
e C (SMC contractile)



f I (injury composite)



g R = I + C (relative state)



h

Primary tests (F2), contractile SMC, $R = I + C$

GSE155468: 7 vs 3 patients
 $g = -0.58 [-1.65, 0.33]$
 exact $P = 0.367$ (120 assignments)

GSE213740: 6 vs 3 patients
 $g = -0.35 [-1.27, 1.33]$
 exact $P = 0.786$ (84 assignments)

Full-pipeline check (control-referenced z, permutation of labels through scoring):
 GSE155468: $P = 0.150$
 GSE213740: $P = 0.048$

All cell-type estimates: limited precision
 (3 controls per dataset; CI from bootstrap).

● GSE155468 (ATAA vs control)
 ■ GSE213740 (TAAD vs control)